

Ushna Malik

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EDUCATION

University of Iowa

Ph.D. in Computer Science

Iowa City, Iowa, USA

Aug 2025 – Present

Lahore University of Management Sciences (LUMS)

B.Sc. in Computer Science; Graduated with Distinction; Dean's Honor List (2021–2025)

Lahore, Pakistan

Aug 2021 – May 2025

RESEARCH EXPERIENCE

Graduate Research Assistant, University of Iowa

SPARTA Lab; advised by *Prof. Rishab Nithyanand*

Aug 2025 – Present

Iowa City, IA, USA

- Auditing whether AI-mediated search exposes opposing political groups to more divergent content than web search: built Python pipelines querying Google Search, GPT, Gemini, and Google AI Overviews with **8 query sets** from pro-choice and pro-life users; measure divergence as within- vs. across-group similarity with bootstrap CIs, permutation tests, and difference-in-differences
- Synthesized **100+ peer-reviewed studies** into a framework linking **9 properties** of AI-generated search answers to **11 user-level consequences** for effort, learning, and trust, grading the evidence behind each link

Research Assistant, LUMS

Internet, Data and Society Lab; advised by *Prof. Zafar Ayyub Qazi, Prof. Ihsan Ayyub Qazi, and Prof. Ayesha Ali*

Sep 2024 – Aug 2025

Lahore, PK

- Studied how physicians prompt, verify, and rely on an LLM during diagnosis: **19 physicians** solved 6 clinical vignettes with ChatGPT-4o (interaction logs plus 12 interviews); built a **4-category taxonomy** of prompting and oversight strategies and showed that **how physicians prompted mattered more than LLM access itself** (62.5% vs. 39.1% diagnostic accuracy for full-case vs. summarised prompts) and that oversight eroded within a session (**42%** drifted from scaffolded to direct queries) (*CHI 2026*)
- Tested whether AI-generated political videos microtargeted to each viewer shift political attitudes, in a preregistered RCT with **197 participants** in Pakistan; co-designed the study, built the pipeline that generated each participant's videos (GPT-4o, ElevenLabs, Midjourney) and the mixed-effects analysis; persuasion rose **+21.8 pp** with **no backfire effect** (manuscript in preparation)
- Tested whether LLM assistance improves AI-trained physicians' diagnostic reasoning, in an RCT with **60 physicians**; built the study platform (vignette delivery, arm-gated LLM access, response logging), recruited and randomized participants, and co-designed the answer keys, grading rubric, and blinded evaluation (*Nature Health 2026*)
- Tested whether AI-trained physicians still over-rely on an LLM when its suggestions are wrong, in an RCT with **44 physicians**; designed the error-seeded ChatGPT-4o suggestions with physicians, extended the platform to both arms, and built the blinded adjudication interface (*NEJM AI 2026*)

PROJECTS

Idiolect Erasure Rate (IER): Measuring Authorship Loss after AI Rewriting | *Python, LUAR*

[GitHub](#)

- Defined IER, the drop in authorship-attribution accuracy after an AI assistant rewrites a message, and measured it with a stylometric SVM and LUAR across 3 corpora, 3 assistants, and 3 rewrite conditions; heavy rewriting cut LUAR attribution by **66.5 pp** on blogs and **52.5 pp** on email but not on topic-bound news, and rewritten text evaded AI-text detectors (*CSCW 2026 workshop*)

Reliability-First LLM Behaviour Audits | *Python, Inspect, Ollama*

[GitHub \(manipulation\)](#) | [GitHub \(eval-awareness\)](#)

- Built two Inspect evaluations (**360** and **150 completions**) testing whether LLMs use dark patterns more toward vulnerable-seeming users and whether they refuse more when a prompt signals a test; validated the LLM judges before scoring, showed that an unvalidated 3B judge manufactures a persona effect that vanishes under a validated 7B judge, and found **67% unsafe compliance** on deceptive product reviews

PUBLICATIONS

Prompting, Oversight, and Adoption: Physicians' Use of Large Language Models for Diagnostic Reasoning in an LMIC

Ushna Malik, L. I. Ahmad, A. Hassan, I. Shafique, E. Mohsin, A. Ali, M. H. Alizai, I. A. Qazi

ACM CHI 2026

The Assistant Erased You: Measuring Loss of Authorship Signals in AI-Mediated Communication

Ushna Malik, M. S. Awan

Broader Impacts of GenAI in Communication Workshop @ ACM CSCW 2026

TEACHING & SERVICE

Graduate Teaching Assistant, University of Iowa

Introduction to Computer Science

Aug 2025 – Present

Teaching Assistant, LUMS

Data Science; Data Structures; Discrete Mathematics; Health Data Science for practising physicians

Aug 2023 – May 2025

Reviewer

BMJ Digital Health & AI; ACM CHI 2027

2026 – 2027

Student Volunteer

ACM CSCW 2026

Oct 2026

RELEVANT COURSEWORK

University of Iowa: Advanced Artificial Intelligence; Human-Computer Interaction; Topics in Social Computing Systems; Computer Security; Algorithms

LUMS: Machine Learning; Topics in Large Language Models; Topics in Internet Research; Probability; Principles and Techniques of Data Science; Databases; Software Engineering

Stanford: High Performance Computing; Deception and Technology

TECHNICAL SKILLS

Languages: Python, R, SQL, C/C++

ML & NLP: PyTorch, Hugging Face Transformers, scikit-learn, pandas, NumPy, statsmodels; text embeddings (Sentence-Transformers, FastText), TF-IDF, authorship attribution, AI-text detection

LLM Evaluation: LLM-as-a-judge design and validation, prompt engineering, Inspect, OpenAI and Gemini APIs, Ollama

Statistics & Research Methods: experimental design, randomized controlled trials (A/B testing), causal inference (DiD), preregistration, mixed-effects models, permutation tests, inter-rater reliability, algorithm auditing, qualitative coding